

E-Commerce Discount Analysis: Studying How Discounts Work on Flipkart and Shopsy

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Abstract

I wanted to understand how discounts actually work on Indian e-commerce websites — are the big discount numbers we see genuine or are they just marketing tricks? To study this I wrote Python scripts to scrape product data from Flipkart and Shopsy across three categories: mobiles, laptops and headphones. After cleaning the data I had 386 products to analyse. I found some really interesting things — headphones show average discounts of 63% which looks suspicious, while mobiles only get about 10% on average. I also built a simple linear regression model to see if I could predict a product's discount from its price and category. The model got an R^2 of 0.69 which I think is decent for real scraped data. The biggest takeaway for me was that a lot of the high discounts on headphones are probably fake — sellers just inflate the MRP to make the deal look better than it is.

Keywords

e-commerce, discount analysis, web scraping, Flipkart, Shopsy, linear regression, Python

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1 Introduction

I shop online quite a bit and I always wonder — when a website says "80% off", is that actually a real discount or is the original price just made up? This question is what motivated me to do this project. I decided to actually collect real data from e-commerce websites and check what's going on with these discounts.

I chose to scrape Flipkart and Shopsy because they are two of the most popular Indian shopping platforms and both have a wide range of products. I picked three categories to study — mobiles, laptops and headphones — mainly because these are the things I personally look at when shopping online and they have very different price ranges which I thought would make the comparison interesting.

The questions I was trying to answer were:

- Which category gets the most discounts and why?
- Is Flipkart or Shopsy cheaper overall?
- Do budget products get more discounts than expensive ones?
- Can a simple model predict what discount a product will get?

I want to be honest that this project had a lot of trial and error. Web scraping turned out to be much harder than I expected because both Flipkart and Amazon kept blocking my requests. I had to switch approaches multiple times before I finally got it working.

2 Data Collection

This was honestly the most challenging part of the project. My original plan was to scrape both Flipkart and Amazon to compare them, but Amazon was nearly impossible to scrape — it loads prices through internal JavaScript calls that even Selenium could not capture. So I switched to comparing Flipkart and Shopsy instead.

2.1 Why Web Scraping Was Difficult

When I first tried using Python's requests library to fetch pages from Flipkart, every single request timed out. I tried changing headers, increasing timeout values, switching to mobile hotspot — nothing worked. After a lot of research I realised the issue was that Flipkart uses JavaScript to render its product listings, so a simple HTTP request only gets you an empty page shell with no product data.

The solution was to use Selenium, which actually opens a real Chrome browser and controls it like a human would. This worked much better because from Flipkart's perspective it just looks like someone browsing normally.

2.2 Tools Used for Scraping

- **Selenium** — opens and controls Chrome browser automatically
- **BeautifulSoup** — reads the HTML and finds the data I need
- **Pandas** — saves everything into CSV files

2.3 How I Found the Right HTML Selectors

One thing I did not expect was that I had to manually inspect the HTML of each website to find the CSS class names for prices and product names. I used Chrome DevTools (F12) to hover over product elements and note down the class names. The tricky part was that Flipkart uses different card layouts for different categories — headphones used a completely different class name than mobiles, which I only discovered when headphones kept returning 0 results.

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Table 1: Summary of Data Collection

Platform	Mobiles	Laptops	Headphones
Flipkart	104	83	40
Shopsy	102	71	40
Total	206	154	80

2.4 Data Collected

For each product I collected: product name, selling price, MRP, discount percentage shown on website, star rating, category, platform, source URL and timestamp. I also calculated the discount myself using the formula $discount = \frac{MRP - Price}{MRP} \times 100$ so I could cross-check against what the website was showing.

3 Data Cleaning and Preparation

The raw data had quite a few issues that needed fixing before I could analyse anything meaningful.

3.1 Problems I Found in the Raw Data

- 31 products had no MRP value — some Flipkart listings just do not show the original price
- 215 products were missing star ratings, mostly from Shopsy
- Some headphone product names had prices and ratings stuck to them as extra text. For example one product name looked like "TRIGGR Trinity 3 with Fabric Finish...4(1,089)Rs.1,199Rs.3,99970% off" because my scraper accidentally picked up surrounding text
- A few rows had just a number like "3.6" as the product name which turned out to be a rating that got scraped instead of the actual name
- 69 products were missing the discount label shown on the website

3.2 How I Fixed These Issues

For missing MRP values I filled them with the selling price, which means those products effectively have 0% discount. This seemed like a reasonable assumption since if there is no original price shown, we cannot claim any discount exists.

For missing star ratings I used the median rating of the same category to fill in the gaps rather than using the overall median. I felt this was more accurate since headphone ratings tend to be different from laptop ratings.

The messy product names took the most work to fix. I wrote a function using Python's `re` library that looks for rupee symbols or "off" text in the name and cuts everything after that point. I had to rewrite this function three times before it correctly handled all the different patterns I found.

After removing rows with invalid product names (just numbers) the final dataset had 386 rows.

3.3 New Columns I Added

I created three new columns to make the analysis easier:

- **price_band** — Budget (under Rs 10,000), Mid Range (Rs 10,000 to 30,000), Premium (above Rs 30,000)

- **discount_band** — Low (0 to 10%), Medium (10 to 30%), High (above 30%)
- **savings_inr** — actual rupees saved which is MRP minus selling price

4 Exploratory Data Analysis

I used Matplotlib and Seaborn to create seven charts to explore the data.

4.1 Overall Statistics

The average discount across all products was 19.1% but the standard deviation was 19.35% which is almost as large as the mean. This tells me the data is very spread out — some products have almost no discount while others claim to have nearly 90% off. The median discount is only 13.88% which is lower than the mean, confirming that a few very high discount products are pulling the average up.

4.2 The Headphones Finding

The most surprising thing I found was how different the average discounts are across categories. Headphones average 63.4%, laptops 18.7% and mobiles only 10.7%. At first I thought maybe headphones just go on sale more often but when I looked at the actual products in the top 10 most discounted list, all of them were headphones with discounts between 70% and 90%. Products like a headphone with MRP Rs 2,599 selling for Rs 295 (88% off). I think these are cases of MRP inflation where the seller sets an unrealistically high original price just to show a big discount number. This is actually a well known practice in Indian e-commerce.

4.3 Platform Comparison

Flipkart and Shopsy offer nearly identical discounts for laptops (19.1% vs 18.1%) and mobiles (10.6% vs 10.8%). The only big difference is in headphones where Flipkart shows 63% average discount. Shopsy headphone listings mostly did not have MRP values which is why there is no bar for Shopsy headphones in my comparison chart — I could not calculate discounts without MRP.

4.4 Price Band Results

I was expecting premium products to get the biggest discounts but it was actually the opposite. Budget products averaged 39.9% discount, Mid Range only 11.9% and Premium 17.2%. After thinking about it I realised this makes sense because budget headphones dominate the budget category and since headphones have inflated MRPs they pull the budget average way up. So this finding is also connected to the headphone inflation problem.

4.5 Discount and Rating

I made a scatter plot of discount percentage against star rating to check if high discount products tend to have worse ratings. There was no clear pattern at all — products with 80%+ discounts had ratings ranging from 3.5 to 4.7. This means you cannot use a high discount as a signal of low quality or vice versa.

5 Data Visualization and Dashboard

I created interactive charts using Plotly in addition to the static Matplotlib charts.

5.1 Why I Used Plotly

Plotly charts are interactive — you can hover over any data point to see details, click on legend items to show or hide specific groups and zoom into specific areas. I thought this was much more useful than static charts for exploring e-commerce data because you can hover over a product in the scatter plot and immediately see its name, price and discount details.

5.2 Charts Built

The interactive dashboard has five charts:

- (1) Discount distribution by platform — shows that Shopsy products are more concentrated in the 0-20% range while Flipkart has a wider spread
- (2) Average discount by category and platform — grouped bar chart showing the headphone anomaly clearly
- (3) Selling price vs discount scatter plot — each dot is a product and hovering shows full details
- (4) Top 15 most discounted products — all headphones from Flipkart
- (5) Savings in rupees by category box plot — shows laptops save the most money in absolute terms even though headphones have higher percentage discounts

The box plot finding was interesting — even though headphones have 63% average discount, the actual rupee savings are tiny (median around Rs 500) because headphone prices are very low. Laptops with only 19% discount actually save you Rs 10,000 to 20,000 in absolute terms.

6 Model Building

6.1 What I Was Trying to Do

I wanted to see if I could predict what discount a product would get just by knowing its price and category. I chose Linear Regression because it is simple to understand and explain, and because the relationship between category and discount seemed roughly linear from the EDA.

6.2 How I Set It Up

The category column contains text values so I had to convert them to numbers using Label Encoding. This gave headphones=0, laptops=1 and mobiles=2. I used selling price, MRP and the encoded category as my three input features, and discount percentage as the target variable.

I split the data 80/20 for training and testing, which gave me 308 rows for training and 78 rows for testing.

6.3 Results

An R^2 of 0.69 means the model explains 69% of why discounts vary across products. The MAE of 7% means on average the prediction is off by 7 percentage points. I think this is a reasonable result considering the data has a lot of noise, especially in the headphones category where discounts vary wildly between products.

Table 2: Linear Regression Model Results

Metric	Value
Mean Absolute Error	7.00%
R^2 Score	0.6914
Training rows	308
Testing rows	78

6.4 What the Model Learned

The most interesting result was looking at the coefficients. The category coefficient was -19.19 which is by far the largest. Selling price and MRP coefficients were both around 0.001 which is practically zero. This tells me that category is what really determines a product's discount — not how expensive it is. Whether a product is in the budget or premium range matters much less than what type of product it is.

6.5 Testing the Model

I tested the model on three example products:

- Mobile at Rs 15,000 → predicted 14.1% (actual average is 10.7%, reasonable)
- Laptop at Rs 50,000 → predicted 27.7% (slightly high but in range)
- Headphone at Rs 500 → predicted 55.5% (actual average is 63%, close enough)

7 Conclusions

This project taught me a lot about both data analysis and how Indian e-commerce platforms actually work. Here are my main conclusions:

- (1) The high discounts on headphones are almost certainly not genuine — they appear to be cases of MRP inflation where sellers list an unrealistic original price
- (2) For mobiles and laptops, Flipkart and Shopsy offer very similar prices and discounts so platform choice does not matter much for these categories
- (3) Looking at percentage discount alone can be misleading — laptops save you far more money in absolute terms despite having lower percentage discounts
- (4) Product category is the biggest factor determining how much discount a product gets — more important than its price
- (5) Web scraping real data is much harder than I expected — I had to debug selector issues, network blocks and data quality problems throughout the project

7.1 What I Would Do Differently

If I had more time I would scrape data across multiple days to track price changes over time. I would also try to find a way to scrape Amazon since it is the biggest platform. And I would look into whether the products with 80%+ discounts actually sell at those prices or if they are just listed that way.

7.2 GitHub Repository

All code, notebooks and data files are available at:
<https://github.com/sahasraa19/ecommerce-discount-analysis>

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